Xiyuan (Tommy) Guo

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Quantum error correction; quantum LDPC decoding; constrained optimization.

Education

University of Virginia Expected May 2027
B.A. in Computer Science; GPA: **3.73/4.00** *Charlottesville, VA*

- **Coursework:** Graduate Quantum Computing (PHYS 5880, Prof. Israel Klich; in progress).
- **Independent study:** Quantum information (Preskill lectures; Solovay–Kitaev theorem and approximation bounds) and quantum mechanics (Griffiths, Sakurai).

Research Experience

Quantum LDPC Decoding – `qudec`
Independent research and open-source software *Project code*

- Implemented LP-based and batched PyTorch ADMM decoders, including parity-polytope projection using the two-slice representation.
- Checked small-code behavior against exhaustive minimum-weight baselines and compared ADMM objective values with an LP solver.
- Benchmarked LP decoding on the $[[144, 12, 12]]$ bivariate bicycle code under code-capacity noise and ADMM variants on $[[72, 12, 6]]$ under phenomenological noise; compared with BP+OSD and studied sensitivity to LLR objective weights, ADMM penalties, and OSD sweeps.

Research Assistant – Yale University Feb 2026 – Present
Prof. Mengxia Liu's Group *New Haven, CT*

- Developed a single-image spectro-polarimetric forward model with **81 wavelength bands and four Stokes parameters**.
- Formulated a constrained inverse problem combining spectral sparsity with Lorentz-cone constraints on physically admissible Stokes vectors.
- Implemented a GPU-enabled PyTorch ADMM solver with Sherman–Morrison linear solves and SLURM execution; investigated calibration cross-talk and spectral unmixing using non-negative least squares.

Research Assistant – UVA BAL Lab Apr 2026 – Present
Prof. Xin Cynthia Tong's Group *Charlottesville, VA*

- Developed **PRISM**, a covariance-regularized framework for multivariate longitudinal missing-data imputation, to study variance attenuation in conditional-expectation estimates.
- Implemented the solver in C++17/RcppArmadillo with projected gradient descent, Armijo line search, mean-constraint projection, and first-order stationarity diagnostics.
- Conducted an exploratory three-qubit tomography study using density-matrix projection to enforce positivity and unit trace; compared random-forest imputation with a linear baseline.

Research Experience (Continued)

Research Assistant – University of Virginia Oct 2025 – Present
Prof. Liqin Ke's Group *Charlottesville, VA*

- Studied transition-metal magnetic anisotropy and its relationship to structural motifs using density functional theory.
- Built the schema, API, and interface for **RareEarth DB**, linking a **4,648-record** magnetic-anisotropy dataset to ICSD/Pearson crystallographic structures.
- Developed interactive three-dimensional atomic-structure visualizations with Three.js and JSmol for structural inspection.

Research Intern – UVA Biocomplexity Institute May 2026 – Aug 2026
Prof. Abhijin Adiga's Group *Charlottesville, VA*

- Built a Julia Independent Cascade simulator and trained graph convolutional networks to estimate cascade size from sparse infection observations.
- Developed an LLM workflow prototype with execution-DAG validation and deduplication; reduced DAG node count by **43.8%** in a three-polynomial demonstration.

Open-Source Contributions

- **Stim:** Submitted PR #1099 to the quantum stabilizer circuit simulator; under maintainer review.
- **difi:** Developed a Go-based Git diff terminal interface with editor-aware navigation; distributed through Homebrew Core and featured on Hacker News.
- **sanjaya:** Built an academic-literature extraction pipeline using OpenAlex, Scrapy, and Playwright.

Honors and Awards

- **Double Hoo Research Grant**, University of Virginia ($5,000) 2026
- **Dean's List of Distinguished Students**, University of Virginia
- **USACO Silver Division**, USA Computing Olympiad

Technical Skills

Programming: Python, C/C++, Julia, Rust, Go, TypeScript, R, SQL.
Scientific computing: PyTorch/CUDA, JAX, PyTorch Geometric, scikit-learn, pandas, RcppArmadillo.
Systems and HPC: Linux, SLURM, Git, Docker, Kubernetes, Bash.
Methods: Constrained optimization, quantum error correction, computational imaging, linear algebra, matrix calculus.

Community Service

Volunteer Firefighter, Plainfield Fire Department 2022 – 2024